Examples of Aws Regions Module-1

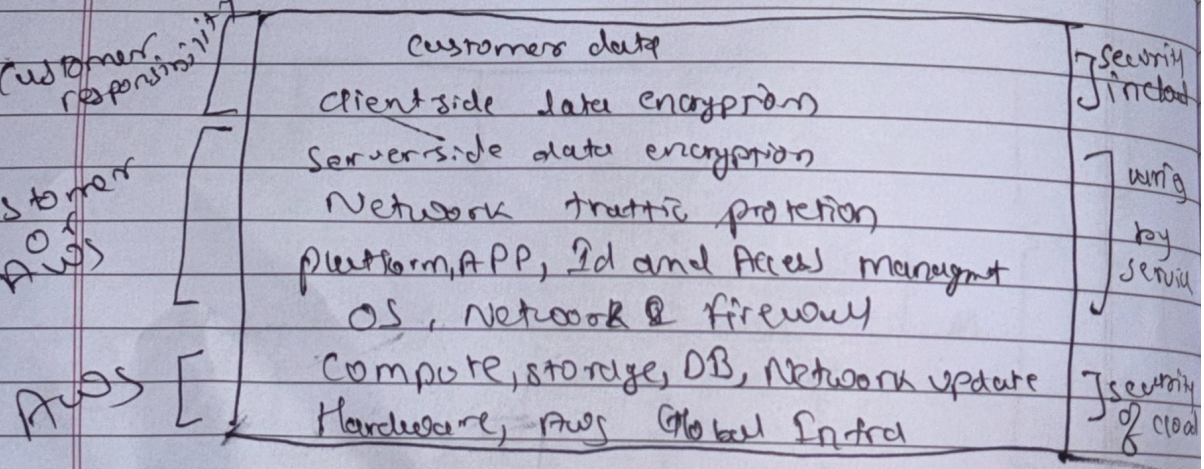
Paris
Tokyo
São Paulo
Dublin
Ohio

Availibility zone

* are not each other to prevent from natural disasters.

Power Networking connectivity

* Aws shared responsibity model



Amazon EC2

Highly flexible
cost-effective
Quick.

Aws built and secured datacentre.
Aws purchased and installed servers.
The servers are online and ready to use.

Multitenancy : Sharing underlying hardware between virtual machines.

Amazon EC2 configurations

- Windows
- Linux.
- Internal business apps
- Webapps
- Database
- Third party software.

Vertical ~~se~~scaling.

Compute as a service (CaaS)

Amazon EC2 Instance families.

General purpose
Compute optimised
Memory optimised
Accelerated Computing
Storage optimised.

General purpose

Balanced resources
Diverse workload
Web seoves
code repositories

Compute optimised

Compute Intensive tasks
Gaming servers
HPC
Scientific modeling

Memory optimised

Memory Intensive tasks

Accelerated

Floating point number calc.
Graphics processing
Data pattern matching
Hardware acceleration

Storage optimised

High performance for locally stored data

Flow:
Instance type
↓
Instance size

API: Application Programming Interface

Interacting with AWS services:
AWS Management console
AWS CLI
AWS SDK.

Using console: Easy to understand:

set up test envs
View AWS bills
View monitoring
Work with non-technical resources

AMI: Amazon machine Image

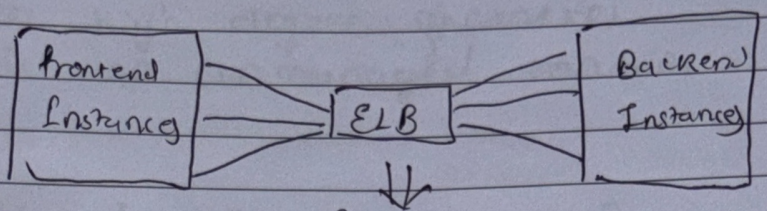
It is a template of the OS & built in application

Ec2 purchase option

on-demand
Saving plans
Reserved instances.
Spot Instances.
Dedicated Hosts.

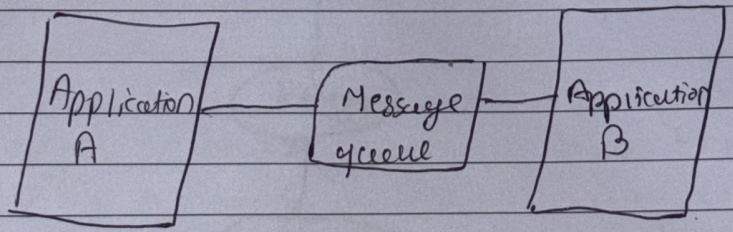
Load balancing : Takes request and groups to instances

Elastic Load balancing (ELB)



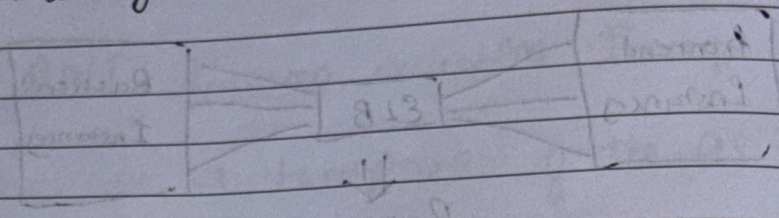
Balances the load to Backend Instances.

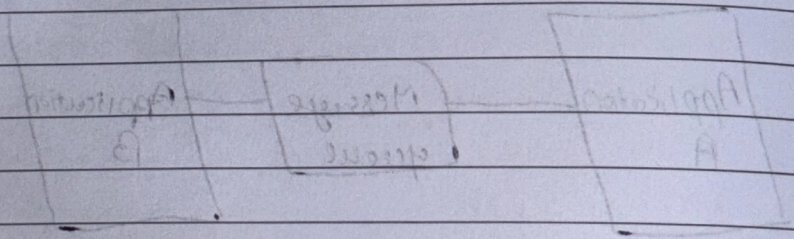
Messaging and queues



Amazon Simple queue service (AWS SQS)
Amazon Simple Notification service (AWS SNS)

payload: data contained within a message





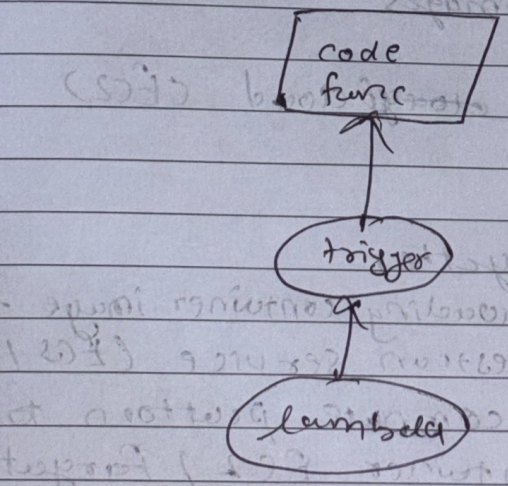
Module - 3.

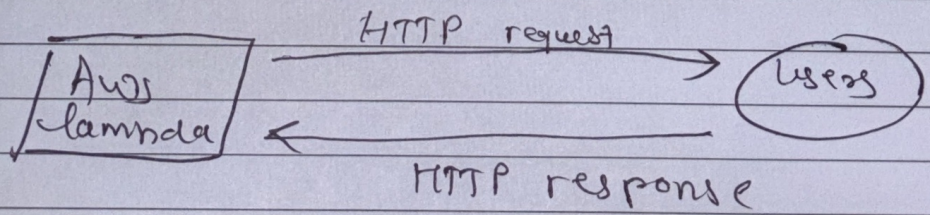
Serverless : You cannot see or access the underlying Infrastructure.

EC2 are Virtual machines.
offers high degree of control
It is an unmanaged control

AWS - Lamda (Serverless)

With lamda you don't need to think about servers or code.





Amazon Elastic container service (ECS)

Streamlined and integrated
Define some parameters
fully managed service.

Amazon Elastic Kubernetes service (EKS)

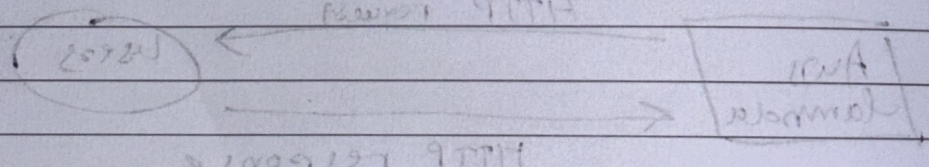
open source platform
More complex
More control and flexibility.

Amazon Elastic container Registry (ECR)

fully managed Docker registry
stores container images

Amazon Elastic compute ~~cloud~~ cloud (EC2)
~~Amazon~~ AWS Fargate.

→ putting it together
start with uploading container image to ECR
Choose an orchestration service (ECS / EKS)
select which compute platform to run your container EC2 / Fargate



AWS Elastic Beanstalk

Simplified provisioning
configuration management

AWS Batch
Intra management
Parallel processing
Auto scaling

Amazon Lightsail
Simplicity
cost-effective solution
Managed infrastructure

AWS outposts
Hybrid cloud solution
consistent envs.
low latency & data residency